

The Joint Embedding Predictive Regularisation of the Cross Sectional Model

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Abstract

This document describes a joint embedding predictive regularisation of a cross sectional multilayer perceptron for monthly portfolio construction. The model retains a selection network that encodes each security, couples the securities through a pooled cross sectional context, and scores them under a maximum Sharpe ratio regression objective, and it adds a self supervised branch that regularises the encoder by asking it to predict the representation of masked characteristics. We develop the full model, present the masking scheme, the predictive objective and its variance and covariance regularisers, the exponential moving average target, and the interpretation of the finding that the self supervised structure does not transfer directly into a higher Sharpe ratio.

1 Overview and motivation

A joint embedding predictive architecture learns representations by predicting the embedding of one part of an input from another, rather than by reconstructing the input in its raw space. Applied to the panel, the idea is that requiring the encoder to infer hidden characteristics from the visible ones should force it to learn the redundancy structure of the characteristics, and that a representation shaped in this way might support a better return signal. The present model adds such a branch to the selection network as a regulariser, controlled by a weight that switches it off at zero and thereby recovers the plain selection network exactly, so that a single specification produces both arms of the comparison.

2 The selection network and the data pipeline

The raw inputs are a panel of monthly characteristics and a list of candidate features, and the target is the excess return of a security over the following month. A characteristic is retained if its coverage is at least ninety percent over the pre test period. For each month every retained characteristic is transformed to its within month percentile rank, missing entries are set to the median before centring so that the inputs lie in the interval $[-0.5, 0.5]$, and securities missing more than one third of the characteristics are dropped. The selection network encodes each security to an embedding, forms a pooled context as the average embedding across the month, and reads a score from a head on the concatenation of the embedding and the context. It is fitted under the loss $(1 - \sum_i w_i r_i)^2$ on a rolling window of at most sixty months, with a fifty epoch cold start, ten epoch warm starts thereafter, three seeds, and seed averaging. The predictive branch shares the encoder with the selection network, so the regularisation acts on the same representation that produces the scores.

3 Structured masking with a visible mask token

Two features of the pretext task are essential to its design. In a training month a subset of the characteristics is chosen at random, and every entry of those characteristics is hidden across all securities. Let $c_j \in \{0, 1\}$ indicate that characteristic j is masked. The masked input replaces the hidden entries by a learned per characteristic value,

$$\tilde{x}_{ij} = (1 - c_j) x_{ij} + c_j \mu_j, \quad (1)$$

where μ_j is a learned mask embedding for characteristic j . The first essential feature is that the masking is visible. Because the inputs are centred ranks about zero, multiplying a masked entry by zero would set it to the median rank, which the encoder could not distinguish from a genuine median value, so the learned mask embedding is used instead to signal that the entry is hidden. The second is that the masking is structured. Whole characteristics are masked across every security rather than individual entries, because the characteristics are heavily redundant and a single hidden entry would be recoverable by interpolation from its correlated siblings, which would make the pretext task trivial. Masking whole characteristics forces genuine inference and mirrors the way characteristics are actually absent in the panel.

4 The predictive objective

The masked input is passed through the online encoder to give $z^c = \text{encoder}(\tilde{x})$, and the unmasked input is passed through a target encoder to give $z^t = \text{encoder}_{\text{target}}(x)$, which receives no gradient. A predictor maps the online representation towards the target, and the invariance loss is the mean squared error

$$\mathcal{L}_{\text{inv}} = \|\text{pred}(z^c) - z^t\|^2. \quad (2)$$

To prevent the representation from collapsing to a constant, two regularisers in the style of variance and covariance regularisation are added. The variance term keeps the standard deviation of each embedding coordinate above a floor,

$$\mathcal{L}_{\text{var}} = \frac{1}{d} \sum_{k=1}^d \max(0, 1 - \hat{\sigma}_k), \quad (3)$$

and the covariance term penalises the off diagonal entries of the empirical covariance of the embeddings,

$$\mathcal{L}_{\text{cov}} = \frac{1}{d} \sum_{k \neq l} \widehat{\text{cov}}_{kl}^2. \quad (4)$$

The branch objective is the sum of the invariance loss and the weighted regularisers, and it is added to the selection loss with a weight λ_{jepa} . The target encoder is updated after each step as an exponential moving average of the online encoder,

$$\theta_{\text{target}} \leftarrow \tau \theta_{\text{target}} + (1 - \tau) \theta, \quad (5)$$

which supplies a slowly moving prediction target and is standard practice for stabilising self supervised training.

5 Interpretation of the empirical result

The evidence on this panel records four successive joint embedding designs, each addressing a limitation identified in the previous one, with Sharpe ratios of 2.58, 2.90, 3.03, and 3.28. The rising sequence reflects improvements to the underlying selection model rather than to the pretext task, because the corresponding controls without the predictive branch rose in step from 3.29 to 3.46, and each design remained below its own control. The final design used genuinely distinct context and target representations with horizon conditioning, so the underperformance cannot readily be attributed to the implementation limitations that were corrected along the way. We can conclude that the structure discovered by the self supervised task, namely the redundancy among the characteristics, does not transfer directly into the return signal on this panel, and that the regularisation, as constructed, is on net a mild drag rather than a benefit.

6 Concluding remarks

The joint embedding predictive model is a careful and well controlled test of whether self supervised representation learning helps portfolio construction on this panel, and the answer here is negative. The value of the model lies in the strength of the comparison rather than in a standalone Sharpe ratio, since the presence of a matched control for every design isolates the effect of the pretext task from the improvements to the selection network. Within the limitations of the present analysis, the finding indicates that shaping the representation to capture the covariation of the characteristics is not the same as shaping it to predict returns, and that on this data the two objectives are not aligned.